SHEAVES ON A QUASI-COMPACT QUASI-SEPARATED SCHEME ARE FILTERED COLIMITS OF CONSTRUCTIBLE SHEAVES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all three parts, the reduction in part (2) to the bounded torsion case being stated and used explicitly.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible sheaf on the small etale site. Same chapter, the definition of constructibility by a partition into locally closed constructible strata on which the sheaf is locally constant. Supplies the objects out of which the colimits are built in all three parts.
- Finite type modules over a Noetherian ring and their colimits. Modules on Sites, the description of a sheaf of modules over a Noetherian ring as a filtered colimit of finite type sheaves of modules. Supplies part (3) and the reduction used for part (2).
- Torsion abelian sheaves and their n-torsion subsheaves. Same chapter, the filtration of a torsion abelian sheaf by its n-torsion subsheaves. Supplies the reduction in part (2) to bounded torsion.

Source reference. Original lemma 093C: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12234–12234. Navigation pointer invoked at proof line 12234; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03S8: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12185–12185. Navigation pointer invoked at proof

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line 12247; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03RZ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11828–11828. Navigation pointer invoked at proof line 12247; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme.*

(1) *Let $\mathcal{F}$ be a sheaf of sets on $X_{étale}$. Then $\mathcal{F}$ is a filtered colimit of constructible sheaves of sets.*
(2) *Let $\mathcal{F}$ be a torsion abelian sheaf on $X_{étale}$. Then $\mathcal{F}$ is a filtered colimit of constructible abelian sheaves.*
(3) *Let Λ be a Noetherian ring and $\mathcal{F}$ a sheaf of Λ-modules on $X_{étale}$. Then $\mathcal{F}$ is a filtered colimit of constructible sheaves of Λ-modules.*

Proof. Let $\mathcal{B}$ be the collection of quasi-compact and quasi-separated objects of $X_{étale}$. By Modules on Sites, Lemma 093C any sheaf of sets is a filtered colimit of sheaves of the form

$$\text{Coequalizer}\Big(\coprod\nolimits_{j=1,\ldots,m} h_{V_j} \rightrightarrows \coprod\nolimits_{i=1,\ldots,n} h_{U_i}\Big)$$

with V_j and U_i quasi-compact and quasi-separated objects of $X_{étale}$. By Lemmas 03S8 and 03RZ these coequalizers are constructible. This proves (1).

Let Λ be a Noetherian ring. By Modules on Sites, Lemma 093C Λ-modules $\mathcal{F}$ is a filtered colimit of modules of the form

$$\text{Coker}\Big(\bigoplus\nolimits_{j=1,\ldots,m} j_{V_j!}\underline{\Lambda}_{V_j} \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} j_{U_i!}\underline{\Lambda}_{U_i}\Big)$$

with V_j and U_i quasi-compact and quasi-separated objects of $X_{étale}$. By Lemmas 03S8 and 03RZ these cokernels are constructible. This proves (3).

Proof of (2). First write $\mathcal{F} = \bigcup \mathcal{F}[n]$ where $\mathcal{F}[n]$ is the n-torsion subsheaf. Then we can view $\mathcal{F}[n]$ as a sheaf of $\mathbf{Z}/n\mathbf{Z}$-modules and apply (3). □